



ROMINA SALJOOGHIAN

AI & Machine Learning Engineer

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PROFILE

AI Engineer with an MSc in Automation and Control Engineering and practical experience building LLM and agentic AI systems, machine learning and computer vision pipelines, and production REST APIs. I take projects from research and model development through to deployment, testing, and live demos. A curious, fast learner and clear communicator who enjoys solving hard problems and collaborates well across disciplines, with a strong focus on shipping reliable, thoroughly tested systems. Seeking AI Engineering roles where my software and control engineering foundation adds real value.

SKILLS & CERTIFICATIONS

Programming	Python (<i>freeCodeCamp</i>) MATLAB SQL (<i>freeCodeCamp</i>)
ML & Deep Learning	PyTorch TensorFlow Keras scikit-learn OpenCV Hugging Face
LLM & Agentic AI	LangGraph (<i>LangChain Academy</i>) LangChain OpenAI API Anthropic API
Backend & API	FastAPI REST API Pydantic pytest
MLOps & Tools	Docker MLflow Hydra Git GitHub Actions Linux L ^A T _E X Automation Explorer (<i>UiPath</i>)
Control & Robotics	ROS2 (Humble) Simulink MPC LQR Kalman Filter PID Kinematics System ID

EDUCATION

MSc · Automation and Control Engineering <i>Politecnico di Milano</i>	Sep 2022 – Mar 2026 Milan, Italy
• GPA 107 / 110 — Thesis: <i>Semantic Characterization of Robotic Environments</i> • Open-vocabulary scene graph generation for robotic manipulation using foundation models (CLIP, SAM, Grounding DINO, RAM) — MERLIN Lab. • Selective Coursework: Neural Networks & Deep Learning · Control of Industrial Robots · Multiagent Systems · Software Engineering · Model Identification · Data-Driven Control · Autonomous Vehicles	
BSc · Electrical Engineering (Minor: Control) <i>K. N. Toosi University of Technology</i>	Sep 2016 – Aug 2020 Tehran, Iran
• GPA 3.65 / 4.0 — Thesis: <i>Classification of Brain Regions in the Mouse via Neural Network</i> • Selective Coursework: Control Systems · Digital Signal Processing · Linear Algebra · Probability & Statistics · PLC Lab	

WORK EXPERIENCE

Delphi Software Developer <i>Torfeh Negar Holding</i>	Jan 2020 – Sep 2021 Tehran, Iran
• Built production UI and backend modules for Holoo accounting software serving thousands of users • Optimized SQL Server queries; resolved bottlenecks and stabilized nightly batch processes	
Instrumentation & Control Intern <i>Tavan Gostare Zino</i>	Jun 2019 – Sep 2019 Tehran, Iran
• Designed and tested industrial grounding and earthing systems for IEC standards compliance	

PROJECTS & RESEARCH

Scene Graph Generation for Robotic Manipulation — Python · PyTorch · CLIP · SAM · GPT API · Hydra 🔗 Live Demo	2025 – 2026
• Designed a modular open-vocabulary pipeline chaining 4 foundation models; 6× improvement over end-to-end baseline (PredCls mR@100: 48.90%) • Built VLM-based annotation pipeline (GPT + Gemini) with structured prompts, producing 246 spatial-relationship triplets for a novel robotics dataset • Zero-shot cross-domain transfer to manipulation scenes (SGDet mR@20: 16.13%) with no fine-tuning • Deployed a CPU-only variant as a ROS2 (Humble) perception node — custom messages, launch files, RViz markers — Dockerised with CI smoke tests; served as a Gradio app on Hugging Face Spaces	
Compliance RAG Agent — Python · FastAPI · LangGraph · ChromaDB · BM25 · OpenAI API 🔗 Live Demo	2026
• Built a RAG backend that indexes regulatory documents and answers questions with source citations via a 4-node LangGraph pipeline • Hybrid retrieval combining vector similarity (ChromaDB) with BM25 keyword search, merged using Reciprocal Rank Fusion; 21 automated tests; deployed on Google Cloud Run with keyless CI/CD	

Visual QA Agent — Python · FastAPI · LangGraph · Anthropic API	2026
• Built agentic backend answering visual questions about images and PDFs through a 4-node LangGraph pipeline with conditional error routing at every stage • Dual-provider VLM abstraction (Claude + GPT-4o); PDF-to-image conversion via PyMuPDF; 11 automated tests with full coverage	
Agentic Job-Application Automation System — Python · Claude · n8n · Task Scheduler · PowerShell	2026
• Built an end-to-end agentic pipeline that ingests a job posting, scores fit, and auto-generates a tailored CV unattended — a headless LLM agent reasoning over a Markdown knowledge base • Orchestrated 3 scheduled workflows (intake, trend recommender, cleanup) with score-gated outputs, idempotent polling, and a git-based review gate; \$0 running cost	
Plant Leaf Disease Classification — Python · TensorFlow · FastAPI · MLflow · pytest	2024
• 5-model ensemble CNN achieving 86% accuracy; async REST API with Grad-CAM explainability endpoint • MLflow experiment tracking with nested runs; 6-module pytest suite covering API, data pipeline, and model registry	
Time-Series Forecasting — Python · TensorFlow · Keras	2024
• Benchmarked 6+ architectures on 48K time series; Conv. LSTM with autoregressive decoding achieved test MSE of 0.01	
CNC Machine — 2-DOF Pantograph Robot — MATLAB · Simulink · Quanser SRV02 Hardware	2024
• Deployed 9 controllers (PID, LQR, MPC, Kalman) on real hardware; MPC: 0.26 s settling, <0.5% overshoot • Near-perfect 2D trajectory tracking with KF + LQR; validated in both time and frequency domains on physical plant	
Networked Control of a Vehicle Platoon — MATLAB	2025
• LMI-based state-feedback controllers for a 4-vehicle platoon under 6 communication topologies; benchmarked centralized vs. distributed cost and convergence trade-offs	

LANGUAGES	
English (Fluent)	Italian (A2) Persian (Native)